



An insight to mobility parameters for AOSTFTs, when the effect of both, localized and free carriers, must be considered to describe the device behavior

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ABSTRACT

The relation between the empirical mobility model parameters used for amorphous thin film transistors and physical device parameters is analyzed, when both localized and free carriers have to be considered to represent the device behavior. A simple procedure is presented to obtain the characteristic temperature and trap concentration of the DOS at the conduction band, using only the linear transfer characteristic at room temperature. An empirical analytical expression that represents analytically the dependence of the surface potential on the gate voltage is presented and validated. Using this expression, the procedure described to calculate the above mentioned parameters is completely analytical. The procedure presented is applied to experimental data of a-IGZO TFTs fabricated with two different technological process.

1. Introduction

The first amorphous oxide semiconductor thin film transistor (AOSTFT) was presented in 2004 by Nomura et al. [1]. It consisted of an amorphous In-Ga-Zn-O (a-IGZO) semiconductor layer, obtained by pulsed layer deposition and a Y_2O_3 layer as the gate dielectric. Indium Tin Oxide (ITO) was used as source (S), drain (D) and gate (G) contacts.

Previously, authors fabricated a single-crystalline $\text{InGaO}_3(\text{ZnO})_5$ TFT [2], for which mobility values above $10 \text{ cm}^2/\text{Vs}$ were obtained. After studying the charge carrier conduction mechanism, they arrived to the conclusion that the carrier transport was associated to percolation conduction over potential barriers around the conduction band edge. These potential barriers are supposed to be due to randomly distributed Ga^{3+} and Zn^{2+} ions in the crystal structure.

In [3], high mobility values were reported for amorphous IGZO (a-IGZO) TFTs, concluding that the percolation conduction mechanism is present, also, in a-IGZO TFTs.

Due to the relatively high electron mobility and high optical transparency obtained with low temperature and relatively low-cost processing techniques, these devices have found important applications, especially in Active Matrix Organic Light Emitting Diodes (AMOLEDs) displays [4].

During the last years, a-IGZO TFTs are object of intensive research, that includes technological and physical aspects. Main objectives are

focused in improving stability, increasing mobility and reducing operating voltage range, among others.

Several publications have studied the characteristics of the a-IGZO band structure and the distribution of localized states DOS, reporting low values of localized states, usually less than $1 \times 10^{20} \text{ cm}^{-3} \text{ eV}^{-1}$, which value strongly depends on the processes used for the device fabrication [5–9].

To consider the possible band conduction in AOSTFTs, authors in [10] proposed the extended mobility edge model, which considers both possibilities, the mobility edge or multiple trapping and release mechanism typical of amorphous TFTs, [11,12], as well as band percolation [13]. The predominant conduction mechanism in AOSTFTs can be due to multiple trapping and release mechanism, conduction in the conduction band including percolation, or a combination of them. This will depend on the characteristics of the fabrication process and operation regime.

To use AOSTFTs in circuit applications, models for these devices, that can be included in the SW tools for circuit designers, are needed. Some models for AOSTFTs presented in literature are based on previous models developed for a-Si:H, to which some specific features of the AOSTFTs have been included [14–16]. This is the case of the Unified Model and Extraction Method (UMEM), developed for a-Si:H TFTs [16], which was complemented with specific features of AOSTFTs, adding also new elements to the extraction procedure.

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In [17], it was also represented the mobility as a power-law dependence of the gate voltage, but they used a different approach, dividing the device operating region into two regions, above and below the threshold voltage V_T . The parameters of the power-law dependence of the mobility in both regions were obtained from the deep and tail localized density of states, which must be known previously.

In [18], another analytical model for current and capacitances in AOSTFTs was developed. In this model, the concentration of both free and trapped charges is considered under some approximations to obtain analytical expressions for solving the Poisson's equation. However, the extraction procedure for the model parameters is complicated and includes material parameters difficult to know with precision. In [19], authors derived two modified power-law dependent mobility, to be used when either trapping or conduction band percolation is predominant.

In this paper, we analyze the relation of the empirical mobility model parameters used in (UMEM) with physical device parameters, when both, localized and free carriers have to be considered to represent the device behavior. For this analysis, numerical calculations without approximations were used to obtain the relation of the mobility model parameters with device physical parameters. An empirical analytical expression, that represents the dependence of the surface potential on the gate voltage, was obtained and validated. A procedure, using this expression is presented to obtain, analytically, the characteristic temperature and trap concentration of the DOS at the conduction band, as well as the gate voltage at which the Fermi level reaches the bottom of the conduction band.

2. Experimental part

The analysis and calculation procedure presented is applied to two-bottom gate top contact a-IGZO TFTs. Device 1, consisted of 150 nm of Si_3N_4 or 200 nm of equivalent oxide, on top of a 12 nm thick a-IGZO layer. Mo/Cr was used for G, D and S contacts. The cross section of the device is shown in Fig. 1. More details of the fabrication can be found in [20].

Device 2, has a similar structure. It consists of 90 nm of HfO_2 , obtained by atomic layer deposition, as gate dielectric and 70 nm thick a-IGZO layer, obtained by pulse layer deposition, as semiconductor layer. Aluminum was used as D and S contacts and 100 nm of Au, deposited on top of 10 nm of Cr, was used as gate contact [21].

The Keithley 4200 semiconductor characterization system was used for the electrical characterizations.

3. Calculation of parameters used in the analysis

To establish a relation between mobility model parameters and device physical parameters, first, the free and localized carrier densities were calculated numerically. The following expression was used for calculating the free carrier density n_{free} using the Fermi-Dirac statistics in terms of the surface potential φ_s :

$$n_{\text{free}}(\varphi_s) = N_C \frac{2}{\sqrt{\pi}} f(\varphi_s), \quad (1)$$

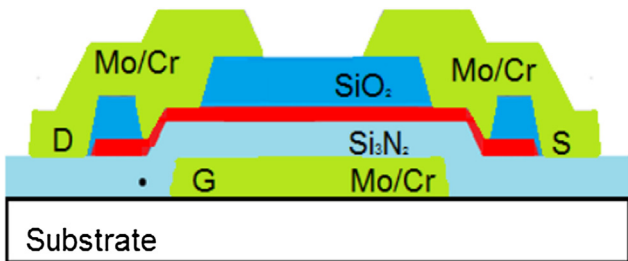


Fig. 1. Cross section of Device 1 a-IGZO TFT.

where

$$f(\varphi_s) = \int_0^\infty \frac{\sqrt{x} dx}{1 + e^{x - \frac{\varphi_s - \varphi_f}{\varphi_t}}}, \quad (2)$$

and $\varphi_t = kT/q$ is the thermal potential at the measurement temperature T in [K]; k is the Boltzmann constant and N_C is the density of states at the bottom of the conduction band E_C , which is taken equal to $5.2 \times 10^{18} \text{ cm}^{-3}$, as reported in [3].

The Fermi level potential φ_f was calculated as:

$$\varphi_f = -\varphi_t \ln \left[\frac{N_D}{N_C} \right], \quad (3)$$

where N_D is the involuntary doping concentration of the semiconductor layer.

If the Boltzmann distribution can be used, Eq. (1) can be approximated to:

$$n_{\text{free}}(\varphi_s) = N_C \cdot e^{\frac{\varphi_s - \varphi_f}{\varphi_t}} \quad (4)$$

For amorphous TFTs, the expression of the localized carrier density, when the characteristic temperature is greater than the room temperature, was defined in [22], that expressed in terms of the surface potential and for the case of a-IGZO TFTs can be written as:

$$n_{\text{loc}}(\varphi_s) = g_{\text{ato}} \varphi_{\text{tt}} e^{\frac{\varphi_s - \varphi_f}{\varphi_{\text{tt}}}} J(\varphi_s), \quad (5)$$

where g_{ato} is the tail trap density at E_C , $\varphi_{\text{tt}} = kT_t/q$ is the characteristic energy of the tail states, and $J(\varphi_s)$ is:

$$J(\varphi_s) = \int_0^{\frac{\varphi_s - \varphi_f}{\varphi_{\text{tt}}}} \frac{1}{1 + z^T} dz. \quad (6)$$

The total carrier density will be the sum of the free and localized carrier density, defined above:

$$n_{\text{total}}(\varphi_s) = n_{\text{free}}(\varphi_s) + n_{\text{loc}}(\varphi_s) \quad (7)$$

From the modeling point of view, it has been previously shown, that UMEM can satisfactorily model AOSTFTs [16]. In this case, the model parameters represent an effective carrier density conformed by both free and localized carriers. We are representing this effective carrier density as:

$$n_{\text{eff}}(\varphi_s) = N_{\text{eff}} e^{\frac{\varphi_s - \varphi_f}{\varphi_{\text{eff}}}}, \quad (8)$$

where N_{eff} is an effective carrier density when $\varphi_s = \varphi_f$ and $\varphi_{\text{eff}} = kT_{\text{eff}}/q$ is an effective characteristic thermal potential.

To obtain N_{eff} , we use two conditions: condition (i) when $\varphi_s = \varphi_f$ and condition (ii) when $\varphi_s = 0$. Substituting (1) and (5) in (7) and considering the condition (i), the following expression is obtained for N_{eff} :

$$N_{\text{eff}} = N_C \frac{2}{\sqrt{\pi}} f(\varphi_f) + g_{\text{ato}} \varphi_{\text{tt}} J(\varphi_f). \quad (9)$$

Equating now (7) and (8) and considering $\varphi_s = 0$, the value for g_{ato} is obtained, using which, the value of N_{eff} can be calculated from (9).

Using the total carrier density expressed through the effective carrier density in (8), the Poisson equation can be written as:

$$\frac{\partial^2 \varphi(x)}{\partial x^2} = -\frac{\rho(x)}{\epsilon_s} = \frac{q}{\epsilon_s} n_{\text{eff}}(x). \quad (10)$$

After solving (10), the surface electric field at the a-IGZO layer is expressed as:

$$E_s = \sqrt{2 \int_0^{\varphi_s} \frac{q}{\epsilon_s} n_{\text{eff}}(\varphi_s) d\varphi_s}, \quad (11)$$

where ϵ_s is the semiconductor permittivity and q is the electron charge.

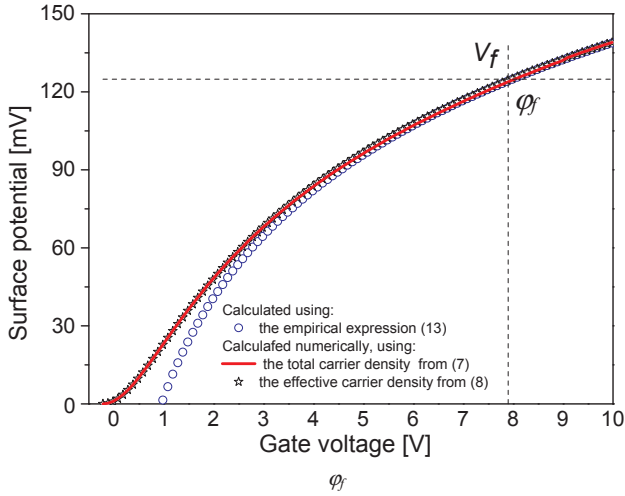


Fig. 2. Surface potential vs. gate voltage calculated using the empirical expression (13), as well as calculated numerically, using the total carrier density (7) and the effective carrier density (8).

After substituting Eq. (8) in (11), solving the integral, and considering Gauss law at the interface, the following transcendental equation is obtained for the relation between V_g and φ_s .

$$V_g - V_t - \varphi_s = \left(\frac{\sqrt{2q\epsilon_s}}{C_{ox}} \right) \sqrt{N_{eff} \varphi_{eff} e^{\frac{-\varphi_f}{\varphi_{eff}}} \left(e^{\frac{\varphi_s}{\varphi_{eff}}} - 1 \right)}, \quad (12)$$

where the C_{ox} is dielectric capacitance per unit area and V_t is the threshold voltage in the case of amorphous TFTs.

To avoid the numerical solution for φ_s as function of V_g in (12), we found the following empirical expression, that provides a very good coincidence with the numerical calculation, see Fig. 2:

$$\varphi_s = 2\varphi_{eff} \ln \left[\frac{C_{ox}}{Q_s} (V_g - V_t) \right]. \quad (13)$$

Q_s is the total charge associated to the effective carrier density, expressed as:

$$Q_s = \sqrt{2q\epsilon_s N_{eff} \varphi_{eff} e^{\frac{-\varphi_f}{\varphi_{eff}}}}. \quad (14)$$

Fig. 2 shows that, the numerical calculation of the surface potential φ_s as function of the gate voltage V_g using the total carrier density from (7) is identical to the one obtained using the effective carrier density from (8). Fig. 2 also shows the calculation of φ_s vs. V_g , using the empirical expression (13) found in this work, where the good approximation with the numerically calculated curves, for a wide range of values of φ_s , is confirmed.

After a detailed analysis, we found that the effective characteristic temperature T_{eff} can be expressed as:

$$T_{eff} = (T + T_i)/2. \quad (15)$$

In amorphous TFTs, the field effect mobility is proportional to the ratio of the free carrier density to the sum of the total carrier density [22]. Using Eq. (4), (7) and (8), the field effect mobility can be expressed as:

$$\mu_{FEc} = \mu_{0c} \left[\frac{n_{free}(\varphi_s)}{n_{tot}(\varphi_s)} \right] = \mu_{0c} \left[\frac{n_{free}(\varphi_s)}{n_{eff}(\varphi_s)} \right] = \mu_{0c} \frac{N_C \cdot e^{\frac{\varphi_s - \varphi_f}{kT_{eff}}} \left(\frac{T_{eff}}{T} \right)}{N_{eff} e^{\frac{\varphi_s - \varphi_f}{kT_{eff}}}}. \quad (16)$$

After some mathematical rearrangement, (16) can be rewritten as:

$$\mu_{FEc} = \mu_{0c} \left[\frac{(A^*)^{\frac{1}{\gamma}} C_{ox}}{Q_0} \right]^{\gamma} (V_g - V_t)^{\gamma}, \quad (17)$$

where μ_{0c} is the band mobility of the device,

$$A^* = \frac{N_C}{N_{eff}}, \quad (18)$$

$$\gamma = 2 \left(\frac{T_{eff}}{T} - 1 \right), \quad (19)$$

$$Q_0 = \sqrt{2q\epsilon_s N_{eff} \varphi_{eff}} \quad (20)$$

From another hand, the expression for the field effect mobility used in the UMEM model is:

$$\mu_{FE} = \frac{\mu_0}{V_{AA}^{\gamma}} (V_g - V_t)^{\gamma} \quad (21)$$

where μ_0 is an empirical value usually considered equal to $1 \text{ cm}^2/\text{Vs}$ for dimensional purposes and V_{AA} is extracted as indicated in [16].

Equating (21) and (17), and substituting the empirical parameter V_{AA} extracted using UMEM by:

$$V_{AAc} = \frac{Q_0}{(A^*)^{\frac{1}{\gamma}} C_{ox}}, \quad (22)$$

a value of μ_{0c} can be obtained and used to calculate $\mu_1 = \frac{\mu_0}{V_{AA}^{\gamma}} = \frac{\mu_{0c}}{V_{AAc}^{\gamma}}$, which is the mobility when $(V_g - V_t) = 1$, independently of the conduction mechanisms present in the device.

The gate voltage, at which the Fermi level reaches the conduction band, V_f can be calculated from (13), (14) and (22), when $\varphi_s = \varphi_f$:

$$V_f = \frac{Q_0}{C_{ox}} + V_t = V_{AAc} A^{*\frac{1}{\gamma}} + V_t \quad (23)$$

The values of g_{ato} , and T_b , which are important material parameters defining the electrical behavior of the device and which are not always known previously, can also be obtained in a relatively simple way.

The procedure for their calculation is described below:

1. Measure a capacitance-voltage characteristic to determine the value of N_D and the dielectric constant of the gate insulator used. Calculate φ_f , using (3). The value of N_C is taken from literature.
2. Calculate the gate capacitance per unit area C_{ox} .
3. From a measured linear transfer characteristic of the device, use the process described in UMEM to extract parameters V_b , γ and V_{AA} as indicated in [16]. These extracted parameters, will consider the effect of all carriers, located and free, present in the device and affecting its electrical behavior. For this reason, they will be associated to the effective concentration of carriers expressed by (8).
4. The values of T_{eff} and T_i are calculated using (19) and (15), respectively.
5. Equating now (7) and (8) and considering $\varphi_s = 0$, obtain the value for g_{ato} .
6. Calculate the value of N_{eff} from (10).
7. Calculate V_{AAc} using (22).
8. Equate (21) and (17) and calculate the value of μ_{0c} .
9. Using (23), calculate the gate voltage V_f for which the Fermi level reaches the bottom of the conduction band.

4. Discussion of results

The results obtained can be used to study the ratio of localized to free carrier in the analyzed devices as function of gate voltage, which is important to understand the electrical behavior of devices. In addition, the procedure described, allows a simple way of calculating g_{ato} , and T_i .

These parameters are usually calculated using other more complicated methods, as the one based in the Meyer-Neldel-Rule (MNR) used in [7,8], which require measurements with temperature, or the multi-frequency Capacitance-Voltage characteristics [12].

As examples, the procedure was applied to two IGZO TFTs fabricated with different techniques were analyzed. Fig. 3 shows the good

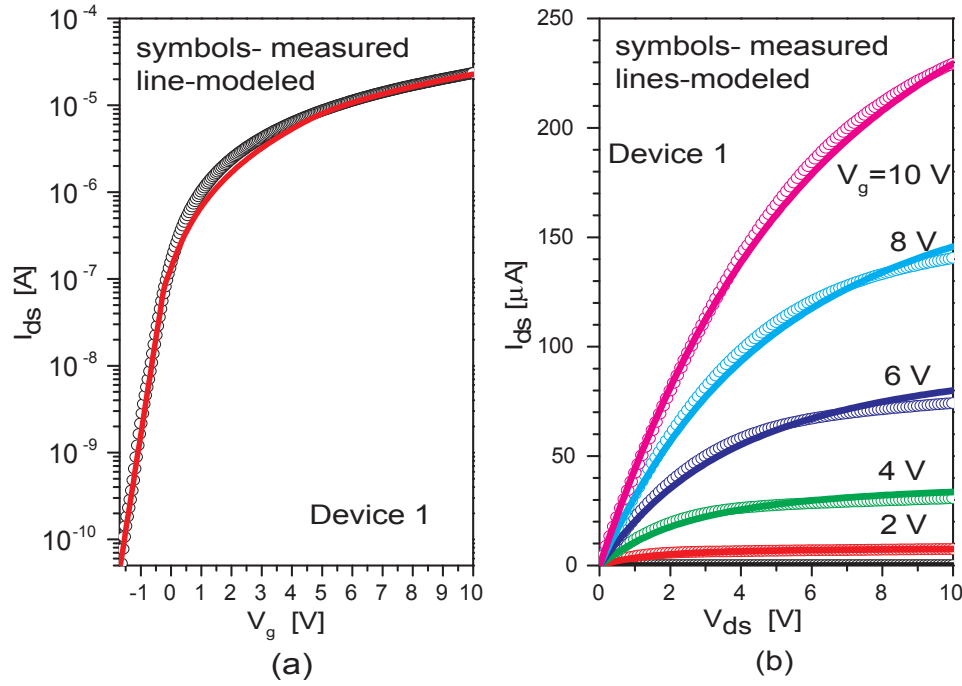


Fig. 3. Measured and modeled (a) linear transfer and (b) output characteristics for Device 1.

agreement between measured and modeled, for the extracted parameters for Device 1 shown on Table 2, used in the analysis.

Figs. 4 and 5 show the good agreement between total carrier density from (7) and the effective carrier density from (8). In the same figures, it can be seen that for these devices, the free charge cannot be considered less than the localized, even when the Fermi level is below the conduction band.

Tables 1 and 2 show the technological and model extracted parameters of the devices, used in the analysis.

Table 3 shows the calculated physical parameters associated to the localized states and the effective carrier density, obtained using the presented procedure.

For Device 1, the obtained value of $g_{ato} = 1.6 \times 10^{19} \text{ cm}^{-3} \text{ eV}^{-1}$ and $T_t = 445 \text{ K}$, which corresponds to a characteristic energy of $E_a = 38 \text{ meV}$. Calculated using the method based in the MNR [7], values of $E_a = 40 \text{ meV}$ and a g_{ato} around $6 \times 10^{18} \text{ cm}^{-3} \text{ eV}^{-1}$ were obtained.

For Device 2, the obtained values N_{eff} and T_{eff} are similar to those of

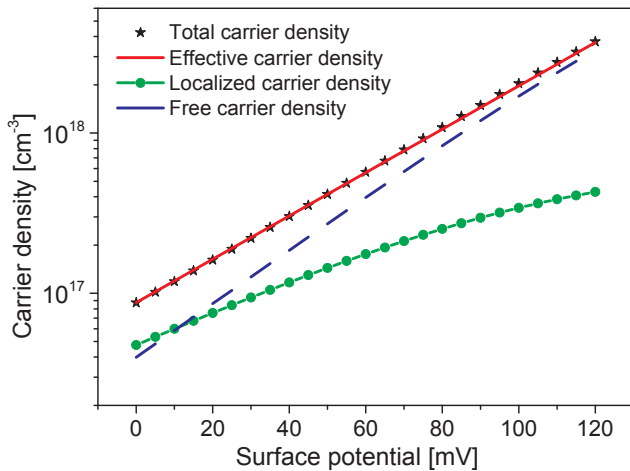


Fig. 4. Total, effective, localized and free carrier densities vs. surface potential for Device 1.

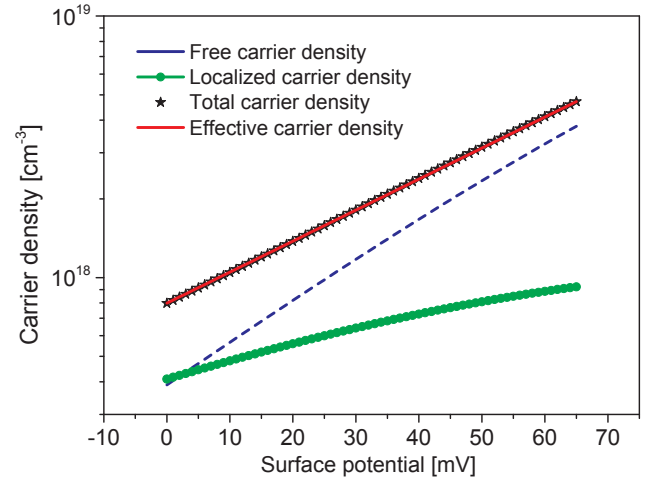


Fig. 5. Total, effective, localized and free carrier densities vs. surface potential for Device 2.

Table 1
Geometrical and physical parameters of the analyzed devices.

Parameter	Device 1	Device 2
W (μm)	900	80
L (μm)	30	40
t_{ox} (nm)	200	90
ϵ_i	5.2	20
ϵ_s	9	9
C_{ox} (nF/cm ²)	23	197
N_D (cm ⁻³)	4×10^{16}	4×10^{17}

Device 1. Since parameter T_{eff} determines parameter γ and the values of μ_1 are also similar for the two devices, the variation of mobility with $(V_g - V_t)$ for both analyzed devices is expected to be quite similar, which is observed in Fig. 6. In addition to the material used as dielectric and its thickness, the other important difference between both devices is the conductivity of the IGZO layer, which in the case of Device 2 is an order

Table 2

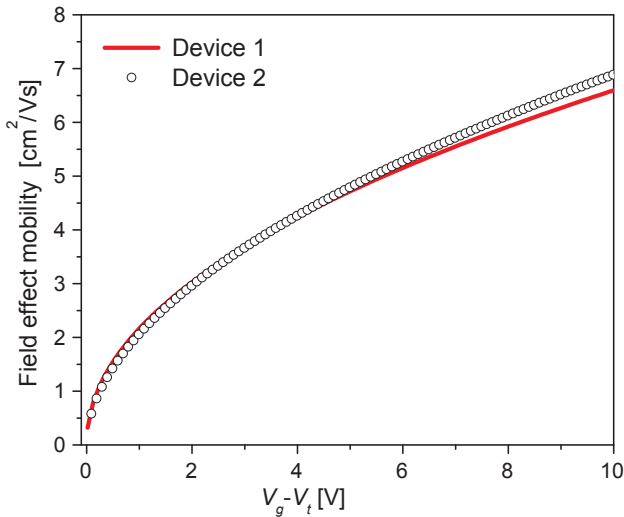
Extracted by UMEM parameters, used in the analysis.

Extracted parameter	Device 1	Device 2
V_i (V)	−0.219	4.61
γ	0.49	0.52
V_{AA} (V)	0.2	0.245

Table 3

Calculated parameters of the TFTs described in the experimental part.

Parameter	Device 1	Device 2
ϕ_f (mV)	125	65
g_{ato} (cm ^{−3})	1.6×10^{19}	1.66×10^{19}
T_i (K)	445	457
T_{eff} (K)	373	378
μ_i (cm ² /Vs)	2.2	2.1
V_f (V)	7.9	5.6
N_{eff} (cm ^{−3})	4.28×10^{18}	4.32×10^{18}
V_{AAc} (V)	5.87	0.73
μ_{oc} (cm ² /Vs)	5.1	1.75

**Fig. 6.** Field effect mobility as function of gate overvoltage ($V_g - V_t$) for both ϕ devices.

higher than in Device 1. For this reason, the gate voltage at which the Fermi level reaches conduction band for Device 2 is lower and the free carrier concentration is always higher than the localized, in spite of the values, for g_{ato} and T_b , being slightly higher than for Device 1.

Also, on Table 3, the values for μ_{oc} and V_{AAc} are different for each device, because although the physical parameters associated to the DOS of both devices are similar, the Fermi potentials are different.

As can be seen, the method presented allows a simple way to calculate g_{ato} and T_i of the a-IGZO layer, as well as to analyze the mobility behavior of different devices and understand the cause of the differences, or similarities, observed.

5. Conclusions

An analysis to relate the amorphous TFTs mobility model with physical device parameters is presented. Calculations allow to determine the relation between free and localized charges in the analyzed devices. In addition, a simple procedure is presented, to determine the localized density of states at conduction band energy, g_{ato} and its characteristic temperature T_b , with the advantage of requiring only a measured linear transfer characteristic at room temperature to which the extraction procedure to model amorphous TFTs previously

developed in the Unified Model and Extraction Method (UMEM) is applied. The procedure developed allows to calculate the relation of the empirical mobility model parameters, which up to now are usually obtained by extraction methods, with device physical parameters. An empirical analytical expression to represent the dependence of the surface potential on the gate voltage is presented and validated. Using this expression, the procedure described to calculate the above mentioned parameters is completely analytical. The procedure also allows to calculate the gate voltage for which the Fermi level reaches the lower edge of the conduction band V_f , in a much simpler way than reported before.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.sse.2018.08.006>.

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